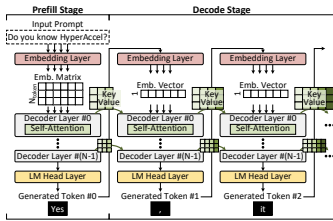


Adelia: A 4nm LLM Processor for Efficient Generative AI Inference

Seungjae Moon, Jung-Hoon Kim, Juntaek Oh, Jay Kim, Joo-Young Kim
HyperAccel, Seoul, South Korea

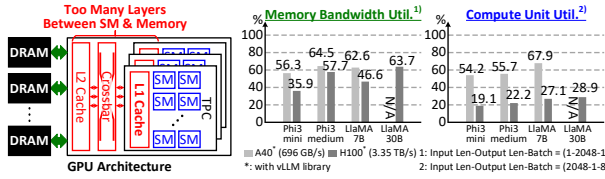
Background & Motivation

LLM Structure



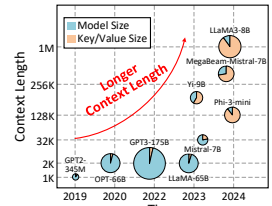
1. Computational imbalance between LLM stages
 - Prefill = GEMM = compute bottleneck
 - Decode = GEMV = memory bottleneck
 - KV cache computed in prefill then continuously appended in decode

GPU Shortcomings



2. Low hardware utilization of GPU
 - Dataflow with complex hierarchy for programmability
 - Consistent data load/store due to little core
 - Massively parallel cores optimized for training, not inference
 - Unnecessary power consumption and high device cost

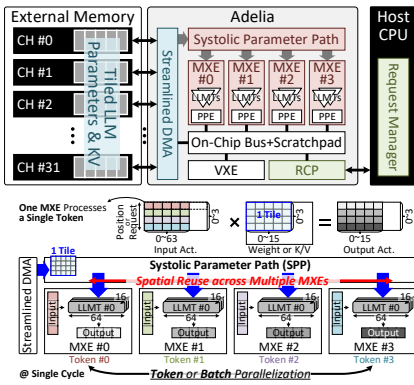
LLM Trend



3. Growth of model parameters and KV cache
 - Increase in computational intensity
 - Demand for scalable hardware
 - Quantization to reduce memory

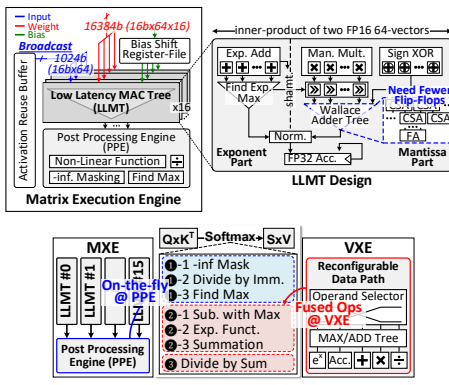
Adelia Architecture

Overall Architecture



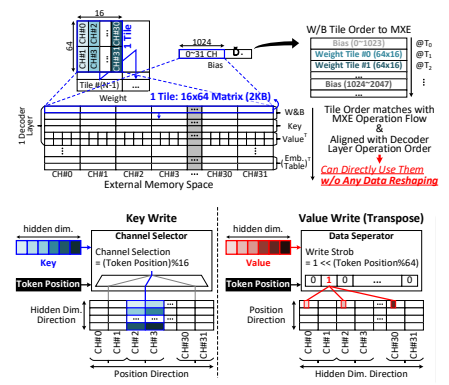
1. Streamlined dataflow for maximum hardware utilization
 - Perfect alignment of memory and compute bandwidth to remove unnecessary buffers
 - Systolic path for spatial reuse of parameters and KV
 - Reduced data load/store due to large core
 - Fully scalable hardware to leverage growing LLM

Microarchitecture



2. LLM-optimized features
 - MAC with hardware sharing to support multi-precision, FP16, BF16, FP8, INT8, INT4, at reduced power and area
 - Fused operations supported by additional processors to hide latency of bottleneck operations

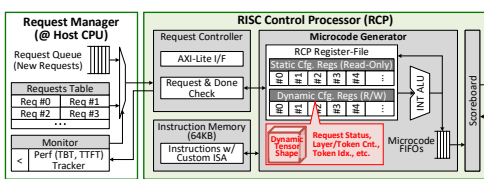
Software-Hardware Co-Design



3. Data preprocessing
 - Hardware-aware mapping of parameters for burst read of data from memory without data reshaping
 - Use of write strobe to handle dynamic movement of activations without physical or performance overhead

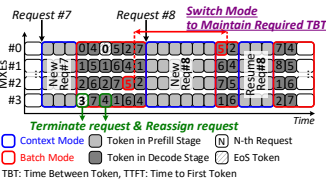
Scheduler

Controller



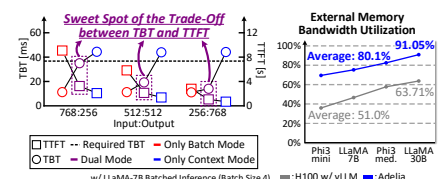
1. Runtime management of user request
 - Continuous monitoring of request information to issue instructions and configuration signals for high efficiency
 - Flexible broadcast of input based on performance requirement

Dual-Mode Parallelization



2. Load balancing based on service-level objective
 - Context mode: prioritize single user for low latency
 - Batch mode: prioritize multi-user for high throughput
 - Support for continuous batching

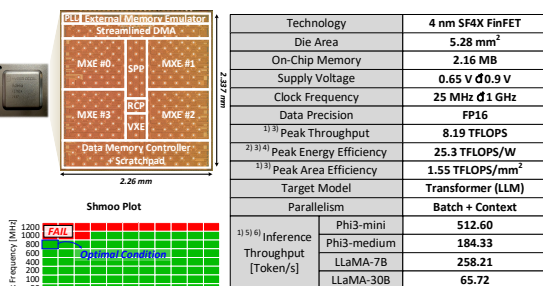
Performance Improvement



3. Achievement of optimal performance and efficiency
 - Ability to configure hardware to meet required TBT
 - Unprecedented memory bandwidth utilization of ~90%

Evaluation

Adelia Specification



Inference Throughput

Model	Batch Size	Token Length [In-Out]	Runtime [s]	Inference Throughput [Tokens/s]
Phi3-mini	1	768-256	2.397	106.79
	2	512-512	4.263	240.17
Phi3-medium	4	256-768	5.993	512.60
	1	768-256	6.902	37.09
LLaMA 7B	2	512-512	12.099	84.63
	4	256-768	16.666	184.33
LLaMA 30B	1	768-256	4.930	51.93
	2	512-512	8.641	118.51
LLaMA 30B	4	256-768	11.897	258.21
	1	768-256	19.778	12.94
LLaMA 30B	2	512-512	34.367	29.80
	4	256-768	46.741	65.72

- At HyperAccel, Adelia is available as a hardware IP that enables LLM in any SoC due to its ultimate scalability.
- Adelia achieves up to x higher normalized throughput than NVIDIA H100.
- Adelia achieves up to x less normalized power consumption than NVIDIA H100.
- With additional improvements, our upcoming single-run, Bertha, will be released as an accelerator card with unprecedented efficiency to reduce the TCO of datacenters when running LLM inference.

Efficiency

